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SECJ 3303 – INTERNET PROGRAMMING

TOPIC 4 – Java Web MVC

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OBJECTIVES

TOPIC 4

Applied

- Use the MVC Pattern to develop your web application, so servlets control the processing and business logic, JSPs handle the presentation and Java POJO (that will convert to JavaBeans in the next topic) for the Model.

Knowledge

- Understand the Model 1 Web Architecture.
- Understand the Model-View-Controller pattern.
- Explain how the MVC design pattern can improve the web application

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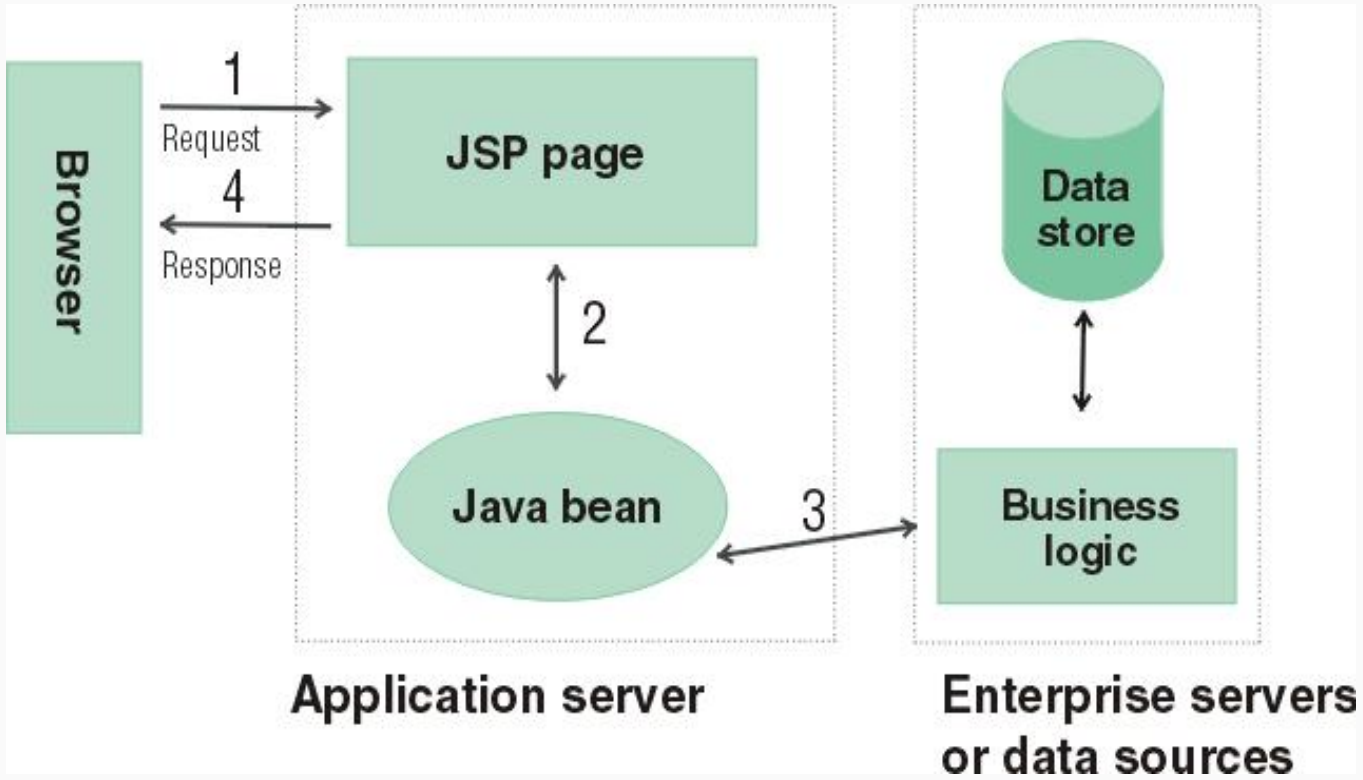
Evolution of Web Application Design Architecture

TOPIC 4

1. Model 1 (Page-centric)
2. MVC Model 2 (Servlet-centric)
3 Web application frameworks
   - Spring
   - Struts

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JSP MODEL 1 ARCHITECTURE



1. The browser sends a request to a JSP page.

1. The JSP page communicates with a Java bean.

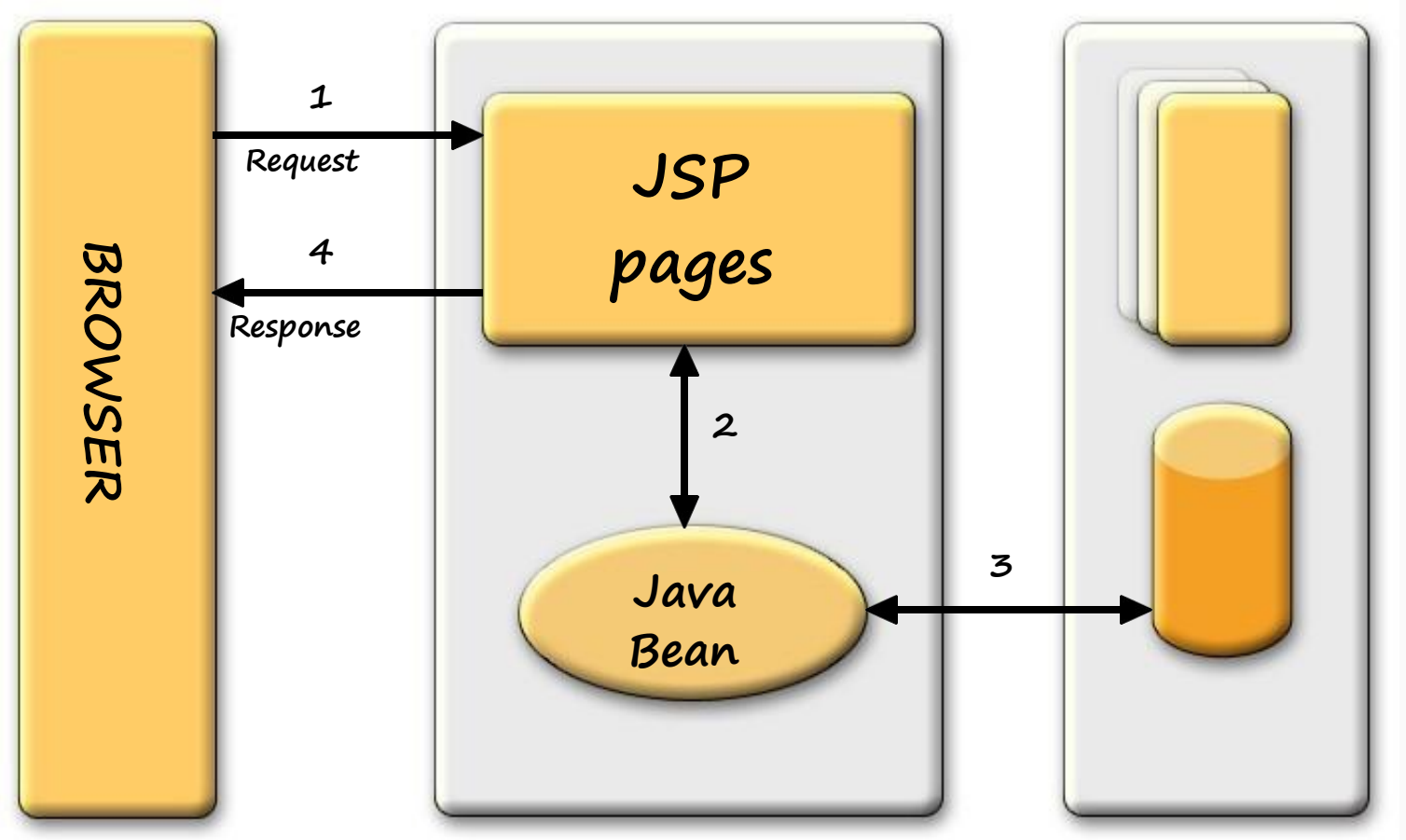
1. The Java bean is connected to a database.

1. The JSP page responds to the browser.

Reference: http://www.setgetweb.com/p/rational/com.ibm.etools.struts.doc_6.0.0/topics/cstrdoc001.html

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Model 1 (Page- centric) Architecture



Page-centric Architecture

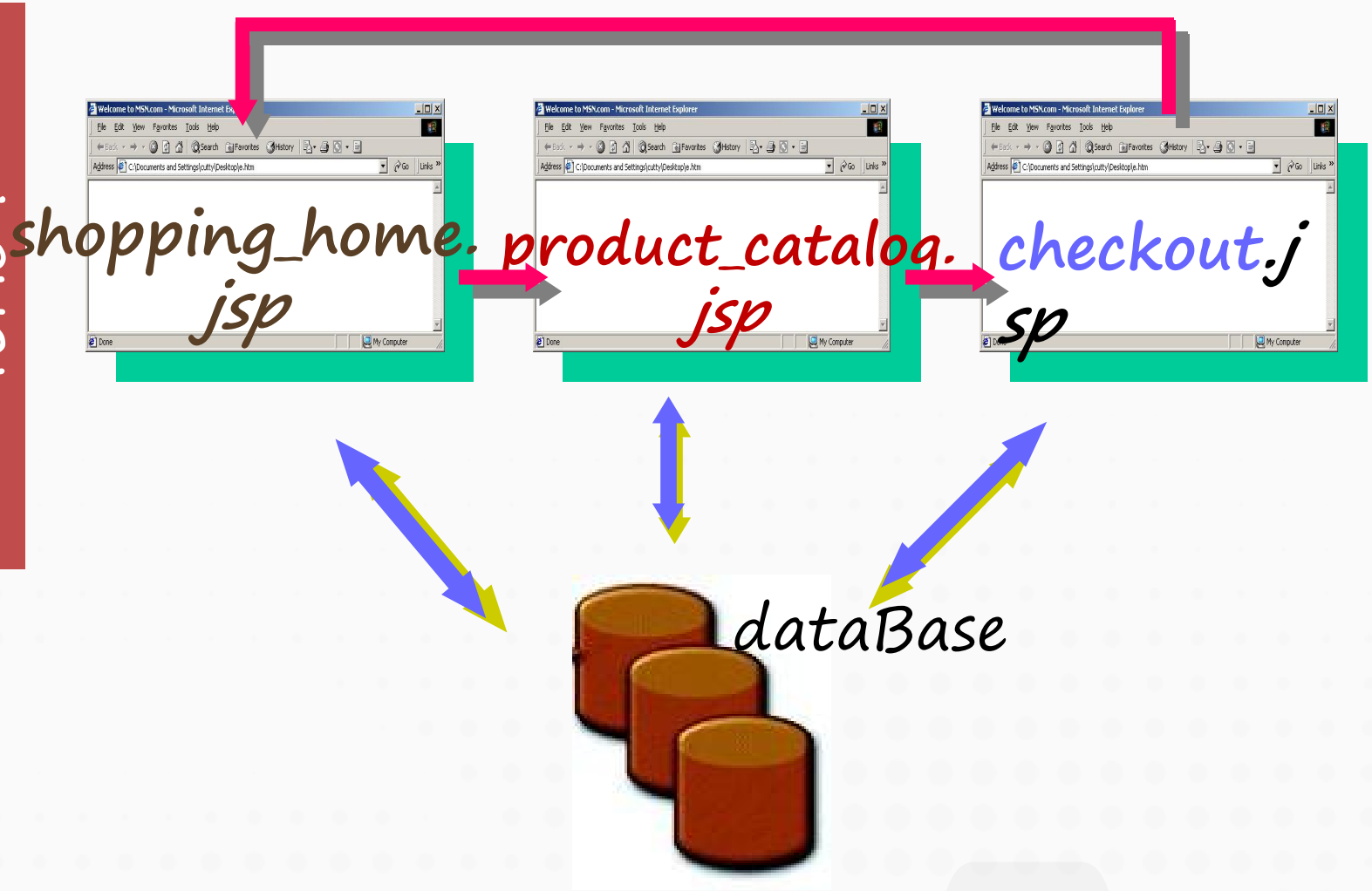
- Composed of a series of interrelated JSP pages
  - JSP pages handle all aspects of the application – presentation, control, and business process
- Business process logic and control decisions are hard coded inside JSP pages
  - in the form of JavaBeans, scriptlets, expression
- Next page selection is determined by
  - A user clicking on a hyper link, e.g. <A HERF="find.jsp>
  - Through the action of submitting a form, e.g. <FORM ACTION="search.jsp">

Page-centric Architecture (cont...)

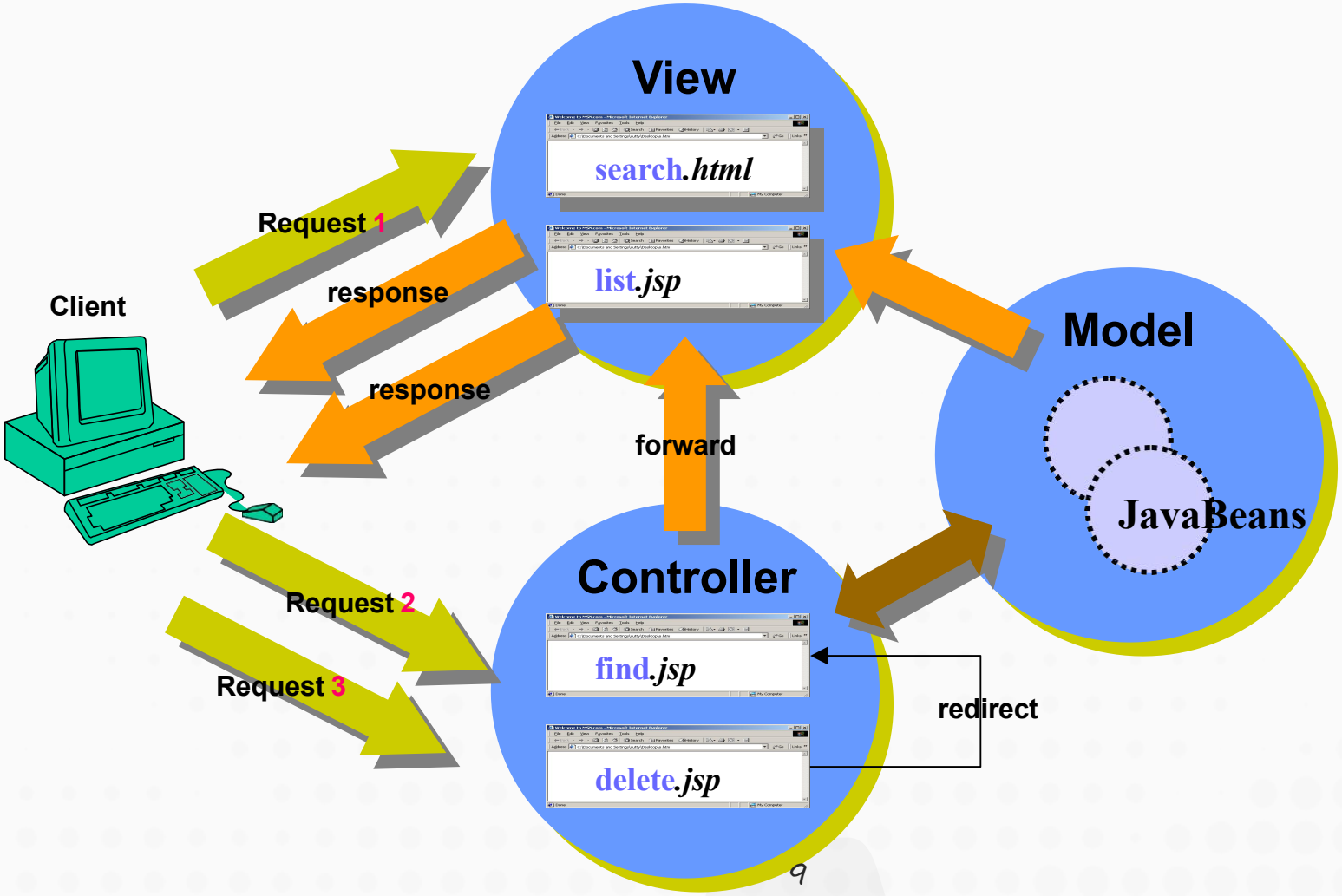
This architecture is sometimes adequate for web application with limited/simple processing requirements

- Generally considered as a bad practice

Page-centric Architecture

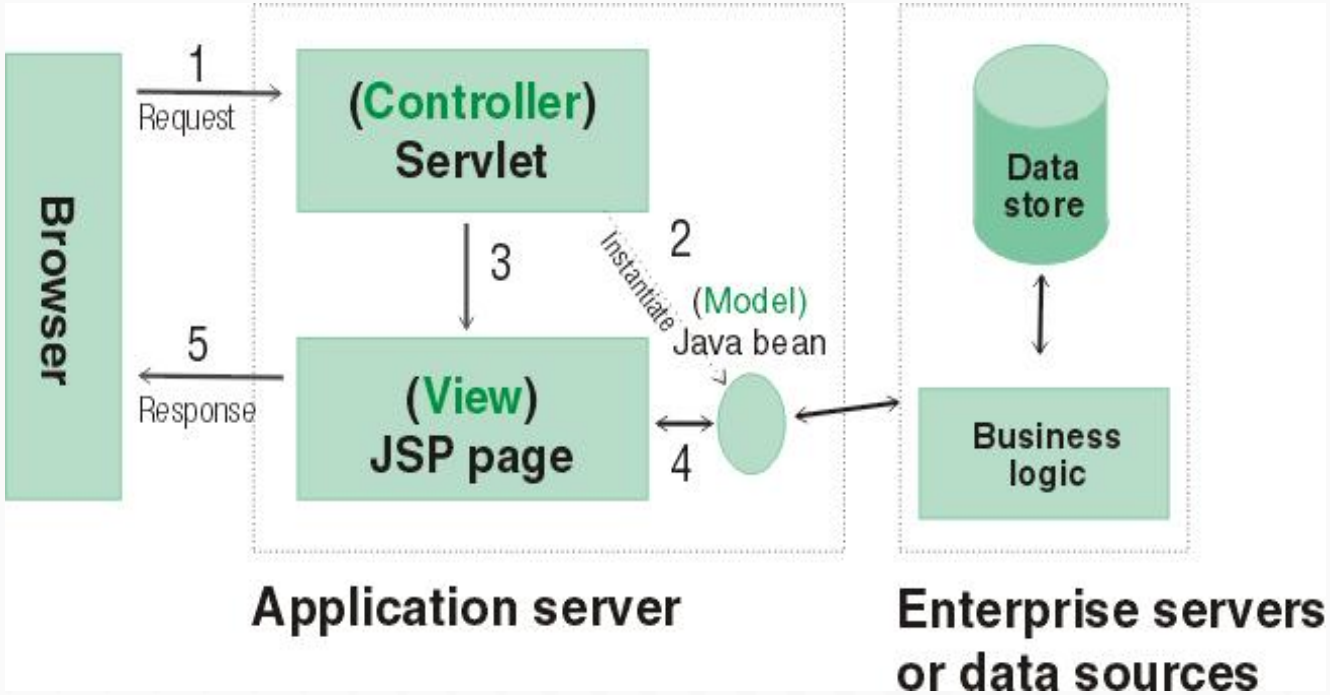


Page-centric Scenario



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JSP MODEL 2 ARCHITECTURE



1.The browser sends a request to a servlet.

1.The servlet instantiates a Java bean that is connected to a database.

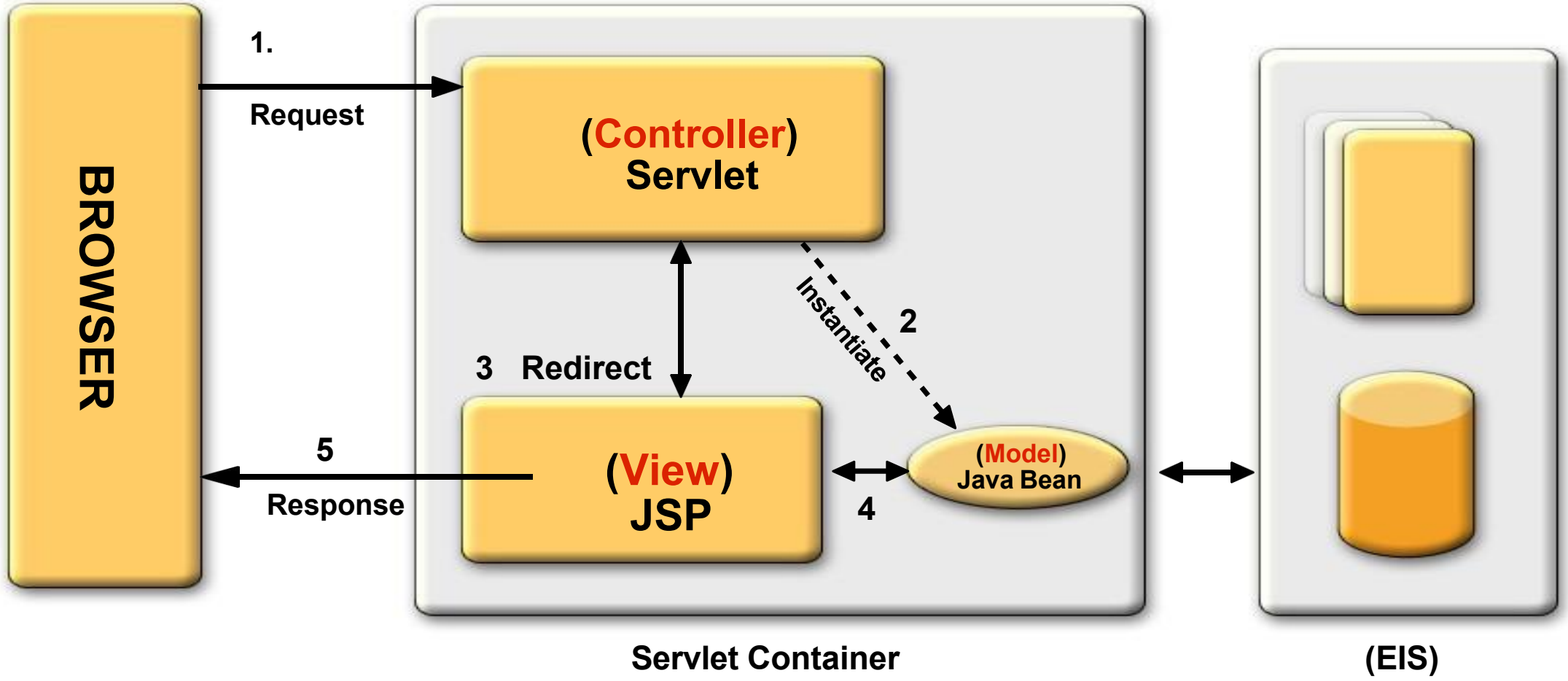
1.The servlet communicates with a JSP page.

1.The JSP page communicates with the Java bean.

1.The JSP page responds to the browser.

Reference: http://www.setgetweb.com/p/rational/com.ibm.etools.struts.doc_6.0.0/topics/cstrdoc001.html

Model 2 (Servlet-Centric MVC) Architecture



Why Model 2 Architecture?

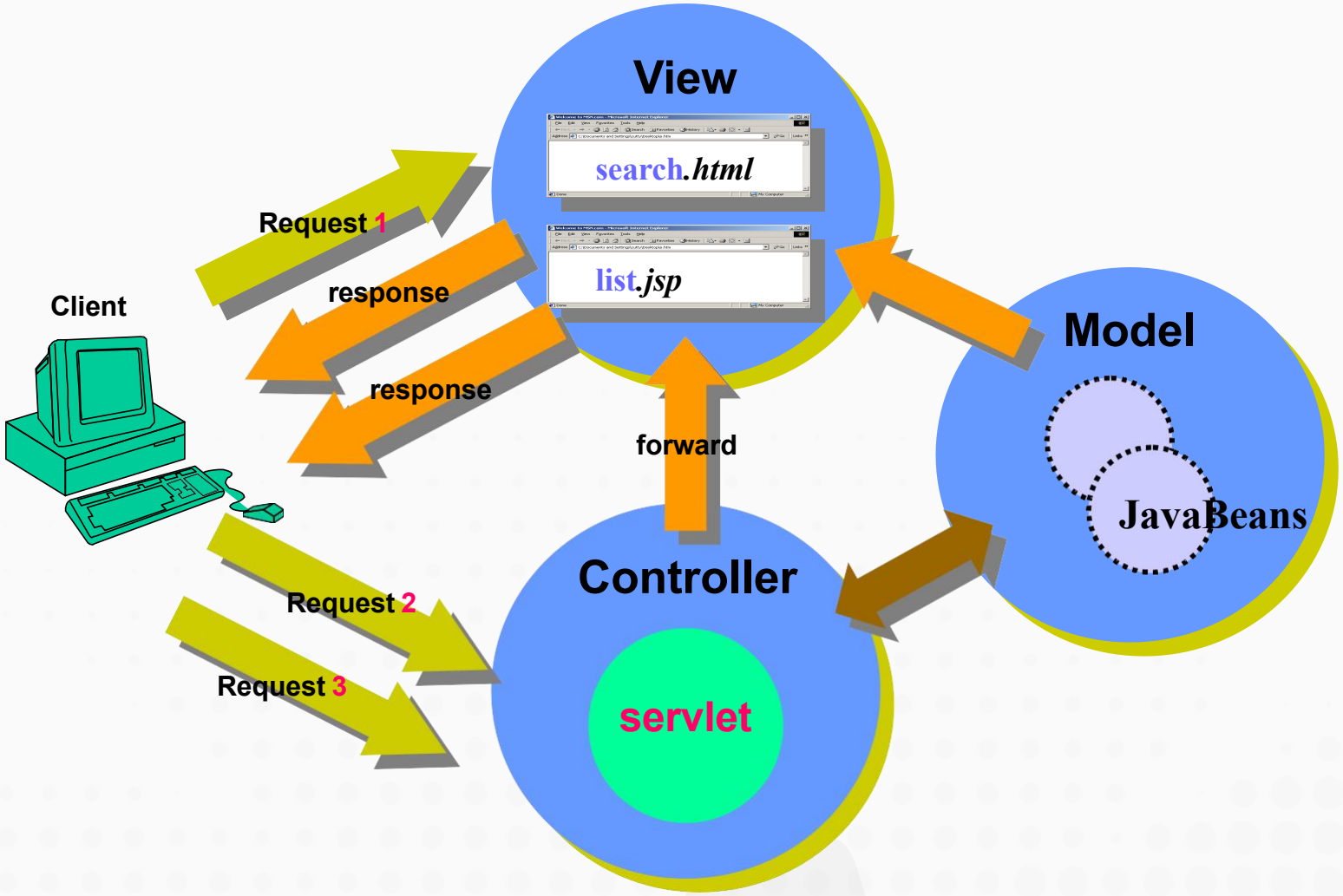
- What if you want to present different JSP pages depending on the data you receive?
  - JSP technology alone, even with JavaBeans and custom tags (Model 1) cannot handle it well
- Solution
  - Use Servlet and JSP together (Model 2)
  - Servlet handles initial request, partially process the data, set up beans, then forward the results to one of a number of different JSP pages

Servlet-centric Architecture

- JSP pages are used only for presentation
  - Control and application logic handled by a servlet (or set of servlets)
- Servlet serves as a gatekeeper
  - Provides common services, such as authentication, authorization, login, error handling, and etc
- Servlet serves as a central controller
  - Act as a state machine or an event dispatcher to decide upon the appropriate logic to handle the request
  - Performs redirecting

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Servlet-centric Scenario



The Three Logical Layers : 1 Model

- Model (Business process layer)
  - Models the data and behavior behind the business process
  - Responsible for actually doing
    - Performing DB queries
    - Calculating the business process
    - Processing orders
  - Encapsulate of data and behavior which are independent of presentation

The Three Logical Layers: 2 View

- View (Presentation layer)
  - Display information according to client types
  - Display result of business logic (Model)
  - Not concerned with how the information was obtained, or from where (since that is the responsibility of Model)

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The Three Logical Layers: 3 Controller

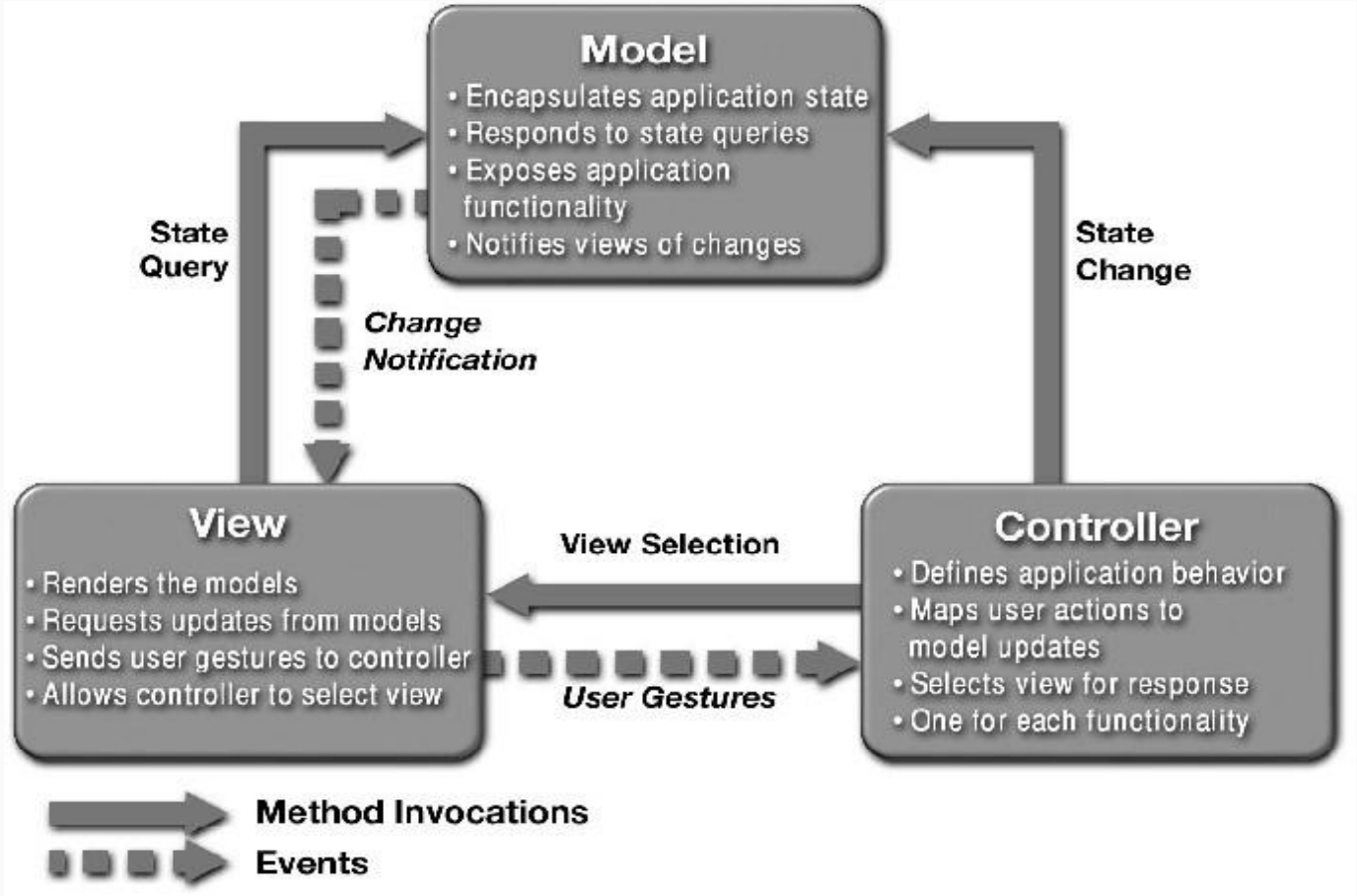
Controller (Control layer)

- Serves as the logical connection between the user's interaction and the business services on the back
- Responsible for making decisions among multiple presentations

  e.g. User's language, locale or access level dictates a different presentation.
- A request enters the application through the control layer, it will decide how the request should be handled and what information should be returned

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MVC Pattern



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Advantages Of Using MVC Framework

1. Clear separation of Data, Presentation and Application Logic

2. Ability To Provide Multiple Views:
   - The model layer returns the same data irrespective of whether the request comes from. The only difference being the controller will choose a different view to render them using an appropriate format or layout

3. The code is much more structured and therefore easier to maintain, test and reuse.
   - Apart from isolating the view from the business logic, the MVC framework reduces complexity for developers modeling an information system, especially when designing large applications.

4. Faster Development Process:
   - it is possible for one programmer to work on the view while others can work on the model and controller to create the business logic of the web application
   - MVC supports rapid and parallel development

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Example - MVC Impl for Login Use Case

components:

LoginController.java :- This is controller part of the application which communicates with model

Auth.java :- The component which contains the business logic for authentication

User.java :- The model which stores username and password for the user

errorPage.jsp :- The view part- If login is not successful then this page is displayed

loginPage.jsp :- The page with input form - username and password

adminPage.jsp :- If login as admin is successful, then this page is displayed

coachPage.jsp :- If login as coach is successful then this page is displayed

traineePage.jsp :- If login as registered trainee is successful, then this page is displayed

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Steps:

Create a new Dynamic web project in STS by clicking

File -> New -> Dynamic Web Project.

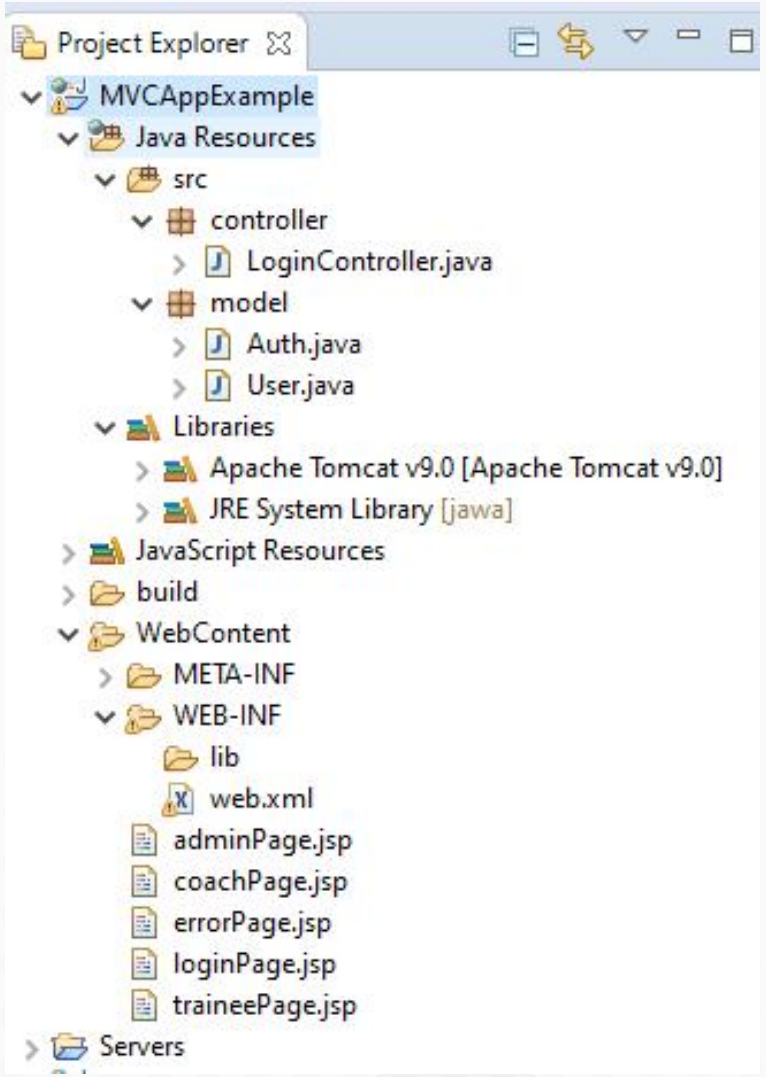
Fill the details i.e. project name, the server. Enter your project name as "MVCAppExample". You will get the following directory structure for the project

Create LoginController.java (controller), Auth.java (model), User.java (model) loginPage.jsp, errorPage.jsp, adminPage.jsp,coachPage.jsp and traineePage.jsp (view) in the corresponding packages

Now that we have file structure, put the following code in corresponding files (to be completed: in-class activity)

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MVCAppExample – Project Directory Structure



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Java Web Application Frameworks

-detail in later topic {Spring Web MVC}

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Web Application Frameworks

- Based on MVC Model 2 architecture
- Web-tier applications share common set of functionality
  - Dispatching HTTP requests
  - Invoking model methods
  - Selecting and assembling views
- Provide classes and interfaces that can be used/extended by developers

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Web Application Frameworks

- Java Spring MVC (will be covered in next topic)

- Ruby on Rails
- PHP Laravel
- CodeIgniter, Etc...

Why Web Application Framework?

- De-coupling of presentation tier and business logic into separate components
- Provides a central point of control
- Provides rich set of features
- Facilitates unit-testing and maintenance
- Availability of compatible tools
- Provides stability
- Enjoys community-supports
- Simplifies internationalization
- Simplifies input validation

Why Web Application Framework?

- Frameworks have evolved with Java Server technology
- JSP/Servlets are still hard to use
- Frameworks define re-usable components to make this job easier.
- A good framework defines how components work to create a usable application.

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the end

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